

A Review of A Multivariate Approach to the Study of Orichalcum ingots from the Underwater Gela's Archaeological Site

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1. Introduction

1.1 Brief description of Orichalcum ingots study Caponetti et al, study

a shipwreck off the coast of Gela, Italy has continued, and continuous, to provide archaeological finds to this day. A discovery dating back to 1980 (Angelucci, 2026), where it was first stumbled upon by divers near Bulala beach. It has been a constant stream of ancient marvels over the decades. We are exploring one of the most significant finds. A veritable hoard of fabled metal; Orichalcum.

Caponetti, Armetta, Brusca, et al. (2017) did a study to determine major & trace elements of the orichalcum ingots. They used this data to find a correlation between the observed different features and the composition.

Our analysis includes the data from their chemometric analysis using ICP-OES and ICP-MS of 38 orichalcum ingots. This is an extension from a fledgling study by the same author, where they performed a less invasive XRF (X-Ray Fluorescence) spectroscopy of the ingots, a non-destructive test that can be performed in the field.

This multivariate analysis does not include the data about physical appearance of the ingots or any correlation analysis of those variables.

Their motivation was to create a starting point for giving insight into archaeometric study of the orichalcum ingots regarding their origin and manufacture technologies.

Our study also has the same motivations. We aim to provide a high level data analysis to provide a detailed, and logical, path to a linkage analysis.

Historically, discoveries of orichalcum have had questionable identification methods. A common

Motivation, problem statement, approach, results, conclusions.

1.2 Table of the Raw data

Table 1: Figure 1: Table 1 The main components and trace elements from Caponetti et al study

Sam-																						
ple	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba	Cu_Zn
S1	73	21	5.600	30580	146.05	5.7	9.85	2.79	12.29	70	0.57	41.4	19.95	55	11.50	3.71	3.17	0.08	1.00	0.19	3.5	
S2	87	16	4.290	53534	199.03	2.06	408.44	20.04	80	1.00	72.8	20.82	174.46	1.55	1.31	0.07	0.15	0.18	5.5			
S3	73	28	5.910	78451	131.00	5.02	3.80	8.47	16.56	70	0.64	29.0	15.74	0.08	8.89	3.33	2.70	0.14	8.90	0.35	2.6	
S4	80	20	6.170	39558	171.00	0.11	104.83	11.99	34	0.82	60.7	23.26	16	13.80	6.08	4.91	0.14	0.29	0.39	3.9		
S5	78	22	3.870	55513	156.07	0.14	506.49	5.3	7.69	0.58	58.7	23.33	87	11.60	4.38	3.69	0.12	0.91	0.45	3.5		

Sam-

ple	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba	Cu_Zn
S6	81	22	6.510.19579	159.00.3	8.32	1.88	9.6	8.54	0.9661.4	15.72.124.36	2.291.900.110.45	0.753.7										
S7	75	20	6.450.33559	130.107.011.00	1.37	11.617.60	0.8051.0	22.31.755.50	2.161.570.130.34	0.503.8												
S8	82	24	6.130.31548	160.075.4	9.66	3.19	7.6	6.98	0.8358.5	23.71.9913.30	0.552.360.111.45	0.263.4										
S9	80	18	4.630.44482	159.109.024.20	5.28	30.53.92	0.9191.3	23.71.698.40	2.701.780.070.11	0.434.4												
S10	72	28	5.920.76435	105.111.023.50	8.29	16.26.65	0.6127.6	19.51.236.77	2.001.410.120.21	0.302.5												
S11	76	19	5.930.43525	138.03.2	15.10	5.13	8.5	8.29	0.8654.5	12.90.926.57	1.711.280.100.56	0.244.1										
S12	69	17	6.670.50657	141.00.1	13.80	1.45	8.3	8.14	0.8755.0	13.41.203.22	1.651.100.103.13	0.224.1										
S13	78	20	3.301.06458	189.049.197.00	160	17.338.30	1.00181.06	61.559.28	3.082.230.070.52	1.174.0												
S14	74	25	4.700.44433	81.7	88.2	14.60	5.08	20.31.98	0.7126.1	29.71.136.51	2.491.550.100.17	0.193.0										
S15	84	20	5.200.14549	171.107.020.60	1.06	23.44.24	0.9592.0	12.00.743.44	1.581.080.080.08	0.174.1												
S16	82	19	4.720.16560	162.103.19.90	1.14	22.24.00	0.8982.1	12.40.6721.20	0.322.530.080.31	0.314.2												
S17	77	20	6.920.79500	177.153.04.70	0.06	31.865.80	0.8127.9	5.8	0.403.31	0.930.670.087.27	0.353.8											
S18	83	19	4.570.44515	158.111.022.10	5.09	28.43.80	0.9989.3	21.50.524.08	1.691.010.0911.80	0.274.4												
S19	71	28	5.250.72494	114.101.08.10	6.14	18.510.40	0.7233.6	17.71.498.36	2.611.540.164.89	3.372.6												
S20	76	19	6.310.41535	139.01.5	14.00	1.41	8.2	8.11	0.9756.3	13.40.622.91	1.340.790.060.42	0.404.1										
S21	74	21	4.880.49423	80.0	68.0	14.40	5.67	22.51.99	0.9125.3	26.30.422.49	0.970.590.070.70	0.233.5										
S22	71	21	6.000.93606	142.175.05.30	15.03	11.842.00	0.5646.3	25.70.8511.60	0.822.190.1215.10	0.613.4												
S23	72	22	6.180.33623	121.07.2	9.12	2.61	11.69.07	0.5840.1	18.30.735.38	2.521.450.110.08	0.143.3											
S24	76	18	6.490.28645	172.02.5	9.43	1.94	9.1	9.75	1.0079.7	14.60.403.76	1.390.860.080.15	0.284.3										
S25	78	18	1.050.75496	162.02.2	56.70	0.34	5.0	4.10	0.9149.8	29.50.615.31	2.021.270.090.17	0.324.3										
S26	75	20	6.350.45543	137.07.1	9.50	4.95	11.28.29	0.8558.6	23.40.4914.60	1.691.100.070.19	0.253.8											
S27	72	20	6.480.41558	140.04.5	9.10	4.34	10.68.19	0.8957.0	18.40.363.05	1.490.850.076.73	0.263.7											
S28	76	21	5.910.83471	172.139.06.50	8.88	54.71.89	0.6627.0	12.50.416.32	2.031.600.094.00	4.593.6												
S29	69	28	6.570.81492	121.122.06.40	7.79	18.89.63	0.7133.3	22.10.535.25	2.741.810.130.93	0.382.5												

Sam-

ple	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba	Cu_Zn
S30	86	16	4.61	0.57	467	196.1	58.3	7.70	10.40	19.84	90	0.99	74.2	22.50	192.16	1.13	0.67	0.07	0.28	0.20	5.4	
S31	77	25	5.25	0.73	400	89.0	101.0	8.40	9.98	22.62	89	0.68	26.3	32.60	313.96	1.80	1.05	0.10	0.14	0.24	3.1	
S32	82	22	7.22	0.72	461	148.0	148.0	2.80	0.05	53.82	84	0.95	39.1	14.00	394.74	2.51	1.55	0.12	0.18	0.31	3.7	
S33	82	20	5.14	0.45	492	166.0	117.0	2.80	5.26	28.53	59	1.00	87.4	21.80	142.02	1.00	0.59	0.06	0.33	0.16	4.2	
S34	80	16	4.48	0.47	468	187.0	132.0	3.80	7.64	18.84	52	1.02	70.4	18.70	142.18	0.89	0.57	0.06	0.09	0.19	5.0	
S35	78	19	6.62	0.24	539	174.0	7.4	9.02	1.91	8.9	9.78	0.99	80.3	14.40	172.63	1.31	0.72	0.08	0.24	0.17	4.2	
S36	78	20	7.71	0.18	683	175.0	4.8	7.04	2.36	6.4	6.57	1.00	62.0	14.20	305.24	2.19	1.97	0.11	0.42	0.23	3.9	
S37	75	23	6.63	0.23	464	131.0	6.0	7.51	3.35	9.7	5.62	0.65	40.6	15.20	102.68	1.23	1.02	0.09	2.88	0.12	3.2	
S38	90	21	2.33	0.38	656	165.0	25.0	201.00	90	24.81	5.70	1.75	151.0	9.10	408.97	3.95	2.34	0.22	2.20	0.42	4.3	

1.3 Our Aim

Why do we care about the problem and the results?

Because it could lead to further impressive archaeological finds

What is the scope of the work and what problem is this paper trying to solve?

Determine whether or not any of the ingots could be grouped. The scope is mapping orichalcum origin.

What was done to solve this problem?

More destructive testing to determine trace elements.

What implications does the answer imply?

Our goal is to create and present a comprehensive, comprehensible, reproducible, and replicable multivariate analysis of the data obtained from the chemometric elemental analysis of 38 orichalcum ingots that were discovered in a 6th Century BC shipwreck off the coast Gela, Italy in 2015.

1.4 Our hypothesis

one of our hypotheses questions is whether our data follows a multivariate normal distribution. probably some hypotheses around the clusterings.

2. EDA

2.1 Brief summary of lab techniques used to gather the the data

Data obtained for the table above containing the main component, trace element concentrations and Cu/Zn weight ratio was obtained by Inductively coupled plasma optical emission spectroscopy (ICP-OES) and Inductively coupled plasma mass spectrometry (ICP-MS) techniques (Caponetti et al). Collecting samples from 38 ancient ingots (they originally collected 40) to analysis the association between the ingots composition.

This is the largest sample of its kind however it is still not a large sample.

For the analysis of our report we will be considering all the main components and trace element concentrations excluding the Cu/Zn ratio as...

2.2 Brief summary of the data

Caponetti, Armetta, Brusca, et al. (2017) determined that the quantity of sample was attributed to error generation. Both samples S38 and S08 had respective quantities 12.3mg and 16.0mg. The average amount of material sampled was reported at ~20mg.

The original study only had access to 38/40 ingots as they were on loan to another institute. This is an extension of that analysis, increasing the sample size at this point of the data analysis would inflate the false positive rate.

We started by performing a simple summary of the data in table 1 and creating a box plot to visualize the data. Something to highlight is that the main elements (Cu, Zn, Pb and Fe) are all in percentages, where as the concentrations of the trace elements are in PPM (parts per a million).

Table 2: Data summary

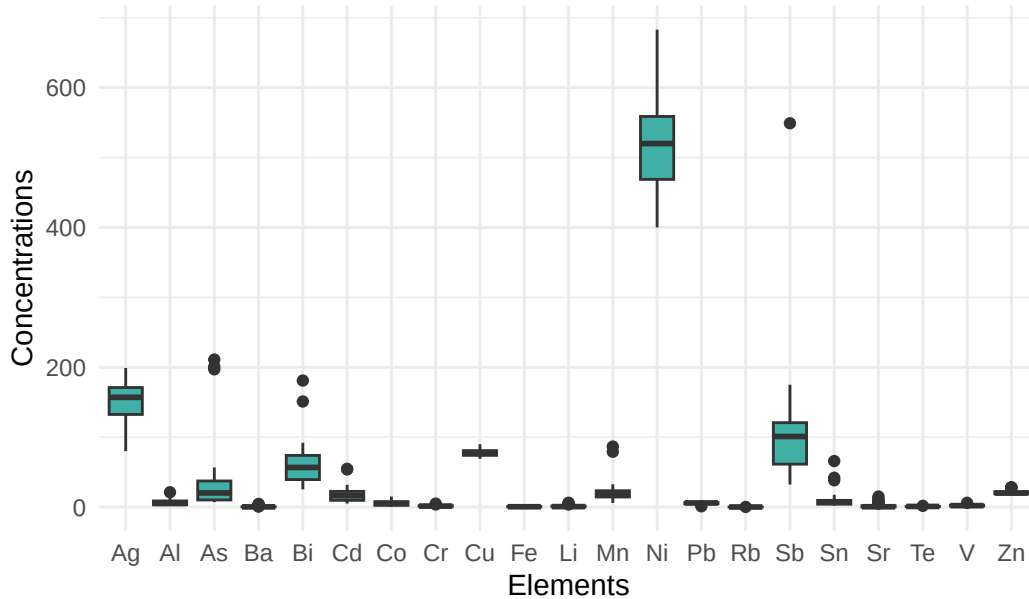
Name	Ingots_F_no[, -1]
Number of rows	38
Number of columns	21
Column type frequency:	
numeric	21
Group variables	None

Variable type: numeric

skim_variable	mean	sd	p0	p25	p50	p75	p100	hist
Cu	77.42	4.98	69.00	74.00	77.00	80.75	90.00	
Zn	20.95	3.25	16.00	19.00	20.00	22.00	28.00	
Pb	5.50	1.34	1.05	4.70	5.92	6.47	7.71	
Fe	0.50	0.23	0.14	0.33	0.44	0.72	1.06	
Ni	525.05	69.56	400.00	468.75	520.00	558.75	683.00	

skim_variable	mean	sd	p0	p25	p50	p75	p100	hist
Ag	149.07	29.59	80.00	132.50	157.00	171.00	199.00	
Sb	105.16	82.28	32.20	61.42	101.00	120.75	549.00	
As	36.81	51.35	7.04	10.16	20.25	37.38	211.00	
Co	5.23	3.25	0.34	2.66	5.02	7.75	15.03	
Cd	17.81	11.38	5.00	9.62	16.35	22.42	54.70	
Sn	10.17	12.43	1.89	4.14	7.34	9.56	65.80	
Te	0.86	0.21	0.56	0.71	0.88	0.98	1.75	
Bi	60.49	32.61	25.30	39.35	56.65	73.85	181.00	
Mn	22.54	15.51	5.80	14.45	19.70	23.37	86.60	
Li	1.23	1.44	0.10	0.40	0.64	1.54	6.16	
Al	6.57	4.29	2.02	3.34	5.28	8.77	21.20	
V	2.23	1.09	0.89	1.50	2.02	2.68	6.08	
Cr	1.60	0.92	0.57	1.01	1.43	1.95	4.91	
Rb	0.10	0.03	0.06	0.07	0.09	0.12	0.22	
Sr	2.05	3.51	0.08	0.18	0.42	2.01	15.10	
Ba	0.59	1.00	0.12	0.21	0.28	0.40	4.59	

Initial boxplot of the table 1 excluding the Cu/Zn ratio



Ni (nickel) has the widest interquartile range, indicating the greatest variation among the samples, while Sb (antimony) shows the largest total range, driven enlarge by sample 13. Unfortunately due to the fact the data has a large range of concentrates (ranging from a minimum of 0.06PPM for Rb and a max of 683PPM for Ni) we will have to consider some transformations to allow for this massive difference in concentrations. looking at the raw data we also noticed some interesting outliers. Particularly, sample 13 for Sb (antimony), which is 549PPM, magnitudes larger than the other samples. We will need to perform transformations on the data as we have a combination of percentages, PPM, major outliers.

But first we will be taking a deeper dive into Copper, Zinc, Lead and Tin as they are the four alloying elements that define orichalcum (probably need a reference for this) we won't be considering Fe due to it being a impurity.

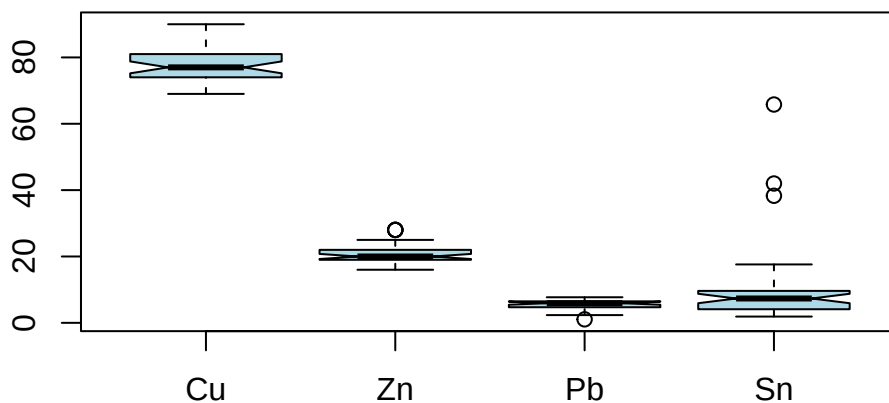
Table 3: Data summary

Name	Ingots_F_no[, c(2, 3, 4, ...
Number of rows	38
Number of columns	4
Column type frequency:	
numeric	4
Group variables	None

Variable type: numeric

skim_variable	mean	sd	p0	p25	p50	p75	p100	hist
Cu	77.42	4.98	69.00	74.00	77.00	80.75	90.00	
Zn	20.95	3.25	16.00	19.00	20.00	22.00	28.00	
Pb	5.50	1.34	1.05	4.70	5.92	6.47	7.71	
Sn	10.17	12.43	1.89	4.14	7.34	9.56	65.80	

Boxplot of Cu, Zn, Pb and Sn (no transformations)



Important to note that Cu, Zn, and Pb are still percentages whereas Sn is PPM, not that there is more tin than lead.

2.2.1 the percentages don't add up

An interesting/surprising observation of the raw data is that the percentages of the main components (Cu, Zn, Pb, and Fe) don't add up to 100% for each of the samples as shown in the table below

Table 4: Table 3: Deeper dive into the main components

Sample	Cu	Zn	Pb	Fe	percentage_total
S1	73	21	5.60	0.30	99.90
S2	87	16	4.29	0.53	107.82
S3	73	28	5.91	0.78	107.69
S4	80	20	6.17	0.39	106.56

Sample	Cu	Zn	Pb	Fe	percentage_total
S5	78	22	3.87	0.55	104.42
S6	81	22	6.51	0.19	109.70
S7	75	20	6.45	0.33	101.78
S8	82	24	6.13	0.31	112.44
S9	80	18	4.63	0.44	103.07
S10	72	28	5.92	0.76	106.68
S11	76	19	5.93	0.43	101.36
S12	69	17	6.67	0.50	93.17
S13	78	20	3.30	1.06	102.36
S14	74	25	4.70	0.44	104.14
S15	84	20	5.20	0.14	109.34
S16	82	19	4.72	0.16	105.88
S17	77	20	6.92	0.79	104.71
S18	83	19	4.57	0.44	107.01
S19	71	28	5.25	0.72	104.97
S20	76	19	6.31	0.41	101.72
S21	74	21	4.88	0.49	100.37
S22	71	21	6.00	0.93	98.93
S23	72	22	6.18	0.33	100.51
S24	76	18	6.49	0.28	100.77
S25	78	18	1.05	0.75	97.80
S26	75	20	6.35	0.45	101.80
S27	72	20	6.48	0.41	98.89
S28	76	21	5.91	0.83	103.74
S29	69	28	6.57	0.81	104.38

Sample	Cu	Zn	Pb	Fe	percentage_total
S30	86	16	4.61	0.57	107.18
S31	77	25	5.25	0.73	107.98
S32	82	22	7.22	0.72	111.94
S33	82	20	5.14	0.45	107.59
S34	80	16	4.48	0.47	100.95
S35	78	19	6.62	0.24	103.86
S36	78	20	7.71	0.18	105.89
S37	75	23	6.63	0.23	104.86
S38	90	21	2.33	0.38	113.71

Table 5: Data summary

Name	main_com[, 6]
Number of rows	38
Number of columns	1
Column type frequency:	
numeric	1
Group variables	None

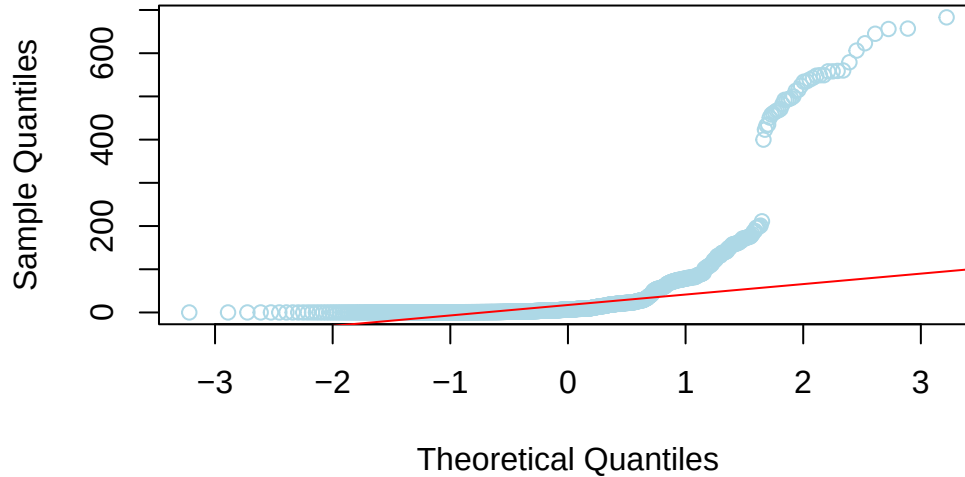
Variable type: numeric

skim_variable	mean	sd	p0	p25	p50	p75	p100	hist
data	104.36	4.28	93.17	101.45	104.4	107.14	113.71	

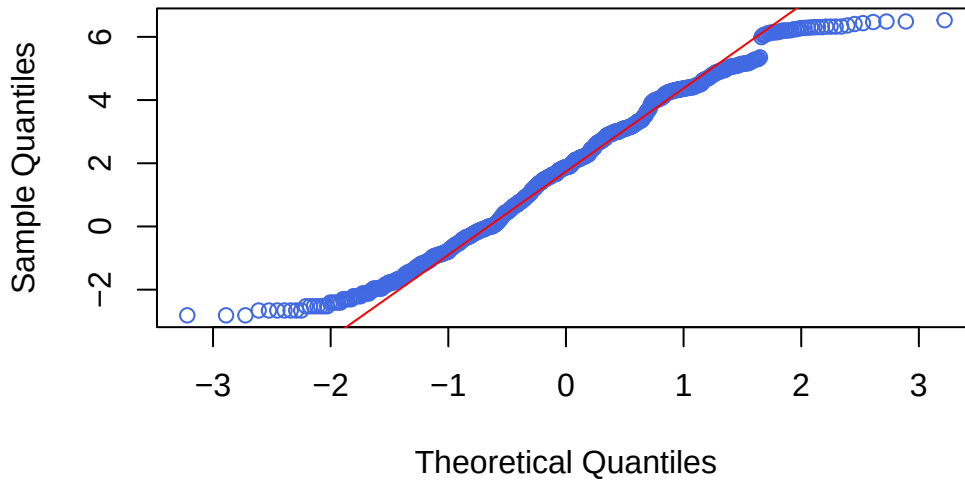
Notice that these samples have a mean of 104.4% with a range 93.17% to 113.71%. Where 33 of the 38 samples exceed over 100% the worse being samples 8 and 38 which the original paper where we got our table 1 data from Caponetti et al. study reasoning it is different from 100 is due to errors in low quantity of the available sample.

2.3 Transformations to start of we had a look at the qq-normal plots of the original data and compared it to the log transformed data also to get idea if normality is met and to help gather evidence if our data follows a multivariate normal distribution.

Normal QQ- Plot of the non-transformed data



Normal QQ– Plot of the log-transformed data



Looking at the QQ-plot with the light blue points (no transformation) we can see some clear deviations in the tails (lower and mostly upper data points), a low amount of points follow the theoretical line (in red), we can assume that our normality assumption wouldn't hold. hence transformations are needed. There are also some slight steps in the points as well as a large jump at 1.7. Performing log transformations on the data produces the QQ-plot with blue points. Notice there are slight deviations in the tails (lower and upper data point), but the rest of the points follow the theoretical line (in red), we can assume that our data normality assumption holds. steps in data and long tails, not happy. Median Absolute Deviation (MAD) may be the answer as it is very robust to outliers. Plus we return a lot of negative values after a log transformation. Either need to add a constant to keep values non-negative or normalize data another way. This is where I start considering removing the massive outliers from the data set.

{this maybe wrong as our data is a combination of percentages and PPM - will need to have

another look at}

2.3.1 Mad Transformation

MAD used because it is more robust to outliers. A log transformation would fix the skew and pull the outliers in, but not eliminate their mathematical influence.

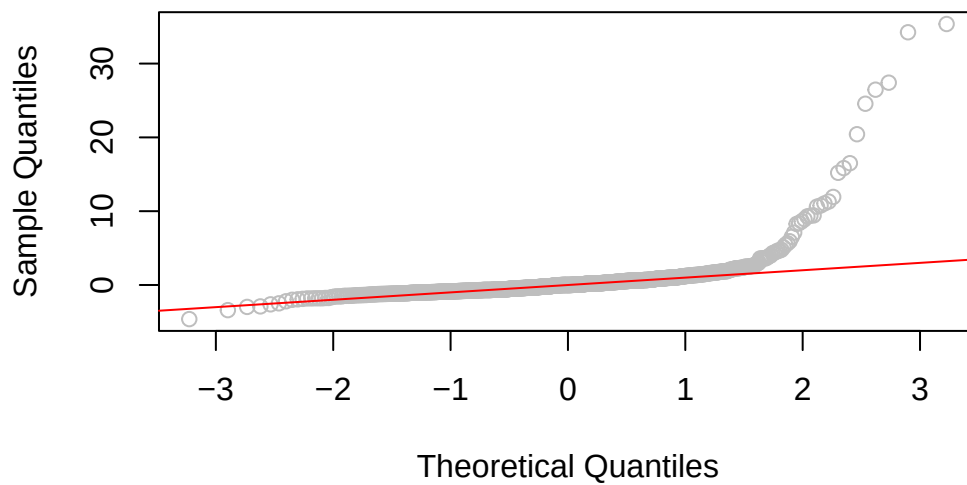
Table 6: Figure something: table of Medians over Mads

	Median	Mad	Median_over_Mad
Cu	77.000	5.189100	14.838797
Zn	20.000	2.223900	8.993210
Pb	5.915	1.052646	5.619173
Fe	0.445	0.207564	2.143917
Ni	520.000	65.975700	7.881690
Ag	157.000	25.945500	6.051146
Sb	101.000	50.260140	2.009545
As	20.250	15.989841	1.266429
Co	5.015	3.914064	1.281277
Cd	16.350	9.562770	1.709756
Sn	7.335	3.691674	1.986903
Te	0.880	0.170499	5.161321
Bi	56.650	26.019630	2.177202
Mn	19.700	6.301050	3.126463
Li	0.645	0.593040	1.087616
Al	5.280	3.409980	1.548396
V	2.025	0.904386	2.239088
Cr	1.430	0.748713	1.909944

	Median	Mad	Median_over_Mad
Rb	0.090	0.029652	3.035208
Sr	0.420	0.415128	1.011736
Ba	0.275	0.126021	2.182176

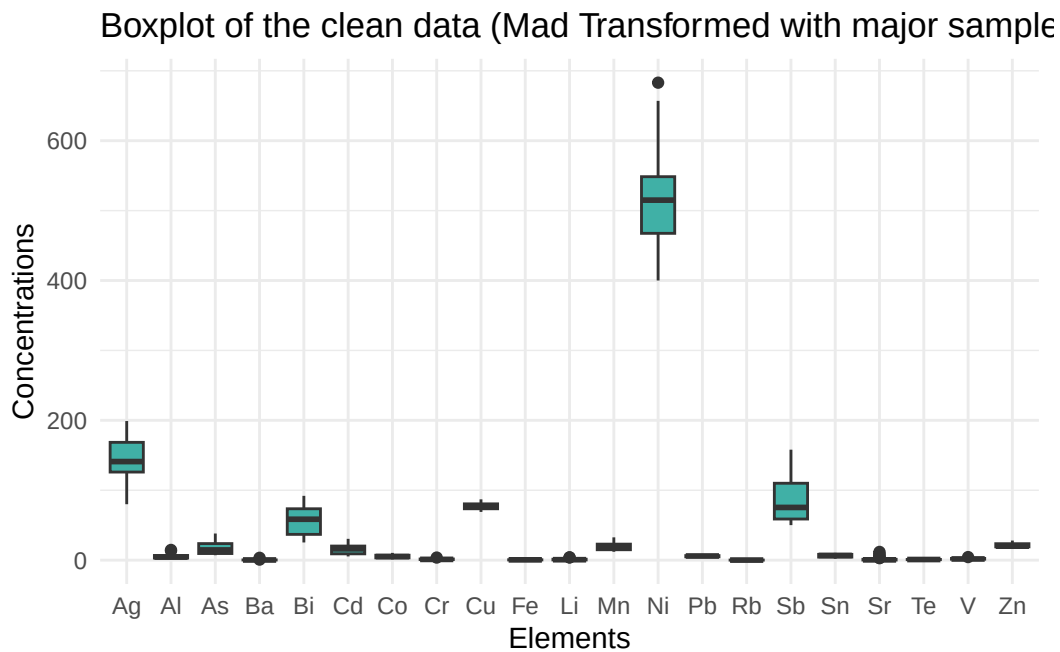
Notice from this table we can see some massive deviations, as expected. Now use the robust Z score to identify outliers that are candidates for removal. leys et al. (2013) argue the MAD is the most robust scale measure under contamination and recommend a median plus or minus 2.5 time MAD rule for outlier detection. .

Normal QQ– Plot of the Mad–transformed data



	Sample	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn
13	S13	78	20	3.30	1.06	458	189	549	197.0	1.60	17.3	38.30	1.00	181.0	86.6
17	S17	77	20	6.92	0.79	500	177	153	54.7	9.06	31.8	65.80	0.81	27.9	5.8

28	S28	76	21	5.91	0.83	471	172	139	46.5	8.88	54.7	1.89	0.66	27.0	12.5
38	S38	90	21	2.33	0.38	656	165	125	201.0	0.90	24.8	15.70	1.75	151.0	79.1
		Li	Al	V	Cr	Rb	Sr	Ba							
13		1.55	9.28	3.08	2.23	0.07	0.52	1.17							
17		0.40	3.31	0.93	0.67	0.08	7.27	0.35							
28		0.41	6.32	2.03	1.60	0.09	4.00	4.59							
38		0.40	8.97	3.95	2.34	0.22	2.20	0.42							

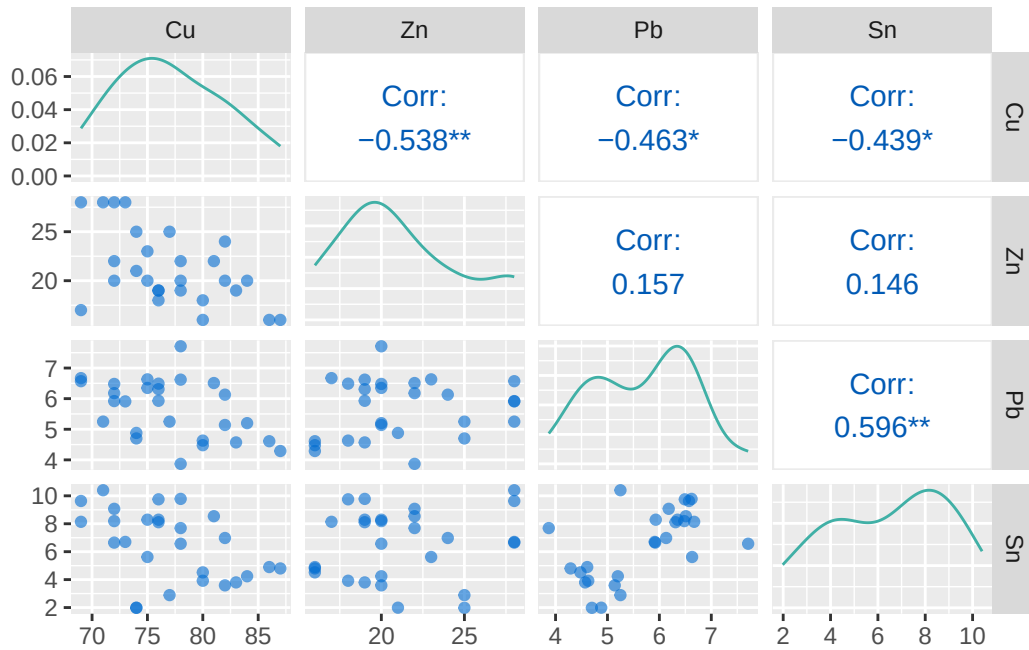


The QQ plot shows very skewed data which we would expected...

What is after we have “cleaned” up the outliers our $|z| > 3$ rule removes samples 13, 25, and 38 which is exactly the sample ingots that Caponetti et al. set aside as their G4 group.

{should probably explain why cutting out 11 samples will be good for the rest of our analysis, something along the line of outliers, terrible samples and low sample quality}

2.3.1 pairwise between CU, Zn, Pb, and Sn using the clean data



Also done to reflect the traditional primary components that make up orichalcum, showing the relationship between these four elements.

2.4 Multivariate normal distribution

What tests did we use? Henze Zirkler

Why did we use those tests? #ALICE STUFF

Henze_Zirkler Test Results:

	Test Statistic	p.value	Method	MVN
1 Henze-Zirkler	1.001	<0.001	asymptotic	Not normal

Henze_Zirkler Test Results:

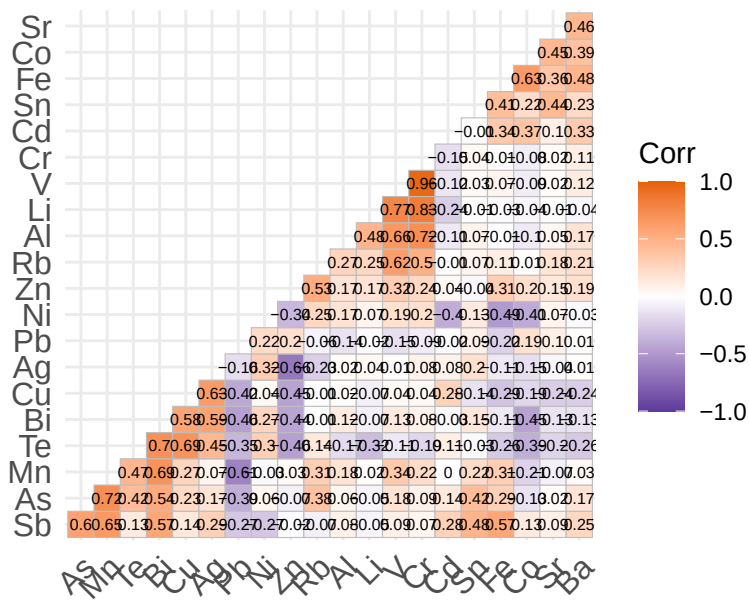
	Test Statistic	p.value	Method	MVN
1 Henze-Zirkler	1	0.046	asymptotic	Not normal

Due to our Henze Zirkler test for both the un-transformed data (p-value <0.001) and the transformed data (p-value 0.046) we can safely reject the null hypotheses and conclude that the data regardless if it is transformed or not does not follow a multivariate normal distribution.

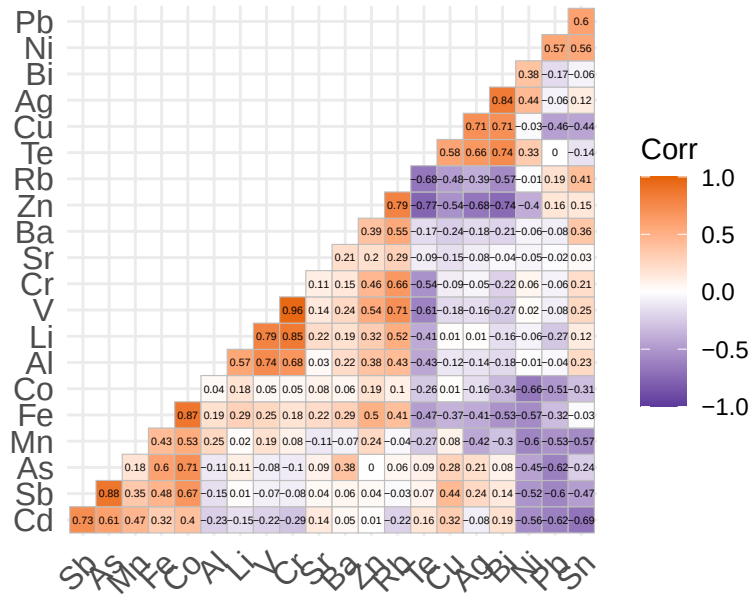
As our data isn't multivariate it isn't appropriate to perform a mahalanobis distance. {could potentially do need to research tho}

2.5 Correlation plots before and after the transformation

Correlation plot of the raw data (no transformations)



Correlation plot of the clean data exluding Cu/Zn



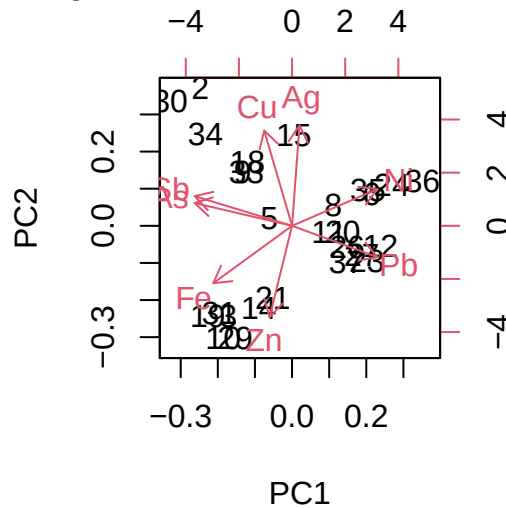
Seeing a lot more correlation in the data with the massive outliers removed. worth keeping and commenting on? Correlation plot some things to highlight; is that the strong positive association between V and Cr for the un-transformed data is also strongly positively association in the transformed data. The transformed data also clearly shows a strong positive association for Bi and Ag which the un-transformed data doesn't show as significantly. the negative association between Zn and Te becomes more stronger in the transformed data.

The transformation didn't just strength some of the associations it also happened to reverse the sign for several of the relationships, Bi and Mn was one the raw data had a positive association but the transformed data turned it into a negative association similar with the relation ships between Sb and Sn as well as Rb and Te. It went the other direction (negative to positive) for As and Co. This is a positive sign that the robustness of the transformation, i.e removing some of the outliers demonstrates their influence on the raw data.

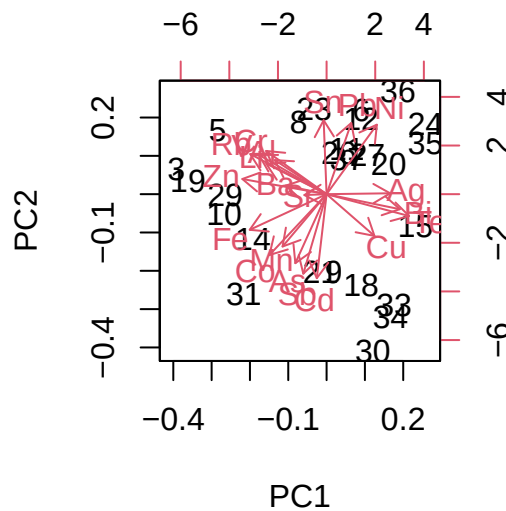
3. LDA and Cluster analysis

3.1 PCA

PCA of only a few elements from the clean data



PCA of all elements in the clean data



unfortunately as our data isn't a multivariate normal distribution it is best that we don't continue performing a PCA analysis and instead focus on a LDA analysis (Vungarala, 2023)

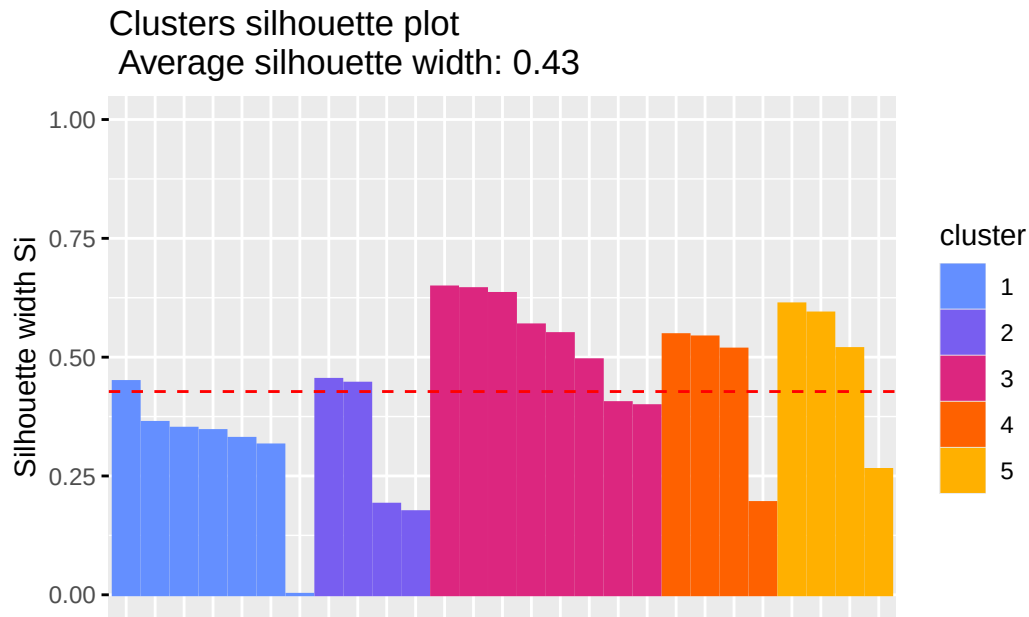
3.2 LDA and PAM we started by performing a PAM (Partitioning Around Medoids) analysis

Table 7: Figure 8: Table contain the k scaled values

k	avg_silhouette
2	0.2034388
3	0.2664019
4	0.2779163
5	0.3101462
6	0.3079801
7	0.2017786
8	0.1778491
9	0.1760813
10	0.1604076
11	0.1586775
12	0.1524900
13	0.1273437
14	0.1047371
15	0.1161656
16	0.1252768
17	0.1504458
18	0.1279253
19	0.1260853
20	0.1158708

k	avg_silhouette
21	0.1067803
22	0.1015029
23	0.0930300
24	0.0610310
25	0.0422777
26	0.0270549

	cluster	size	ave.sil.width
1	1	7	0.31
2	2	4	0.32
3	3	8	0.54
4	4	4	0.45
5	5	4	0.50

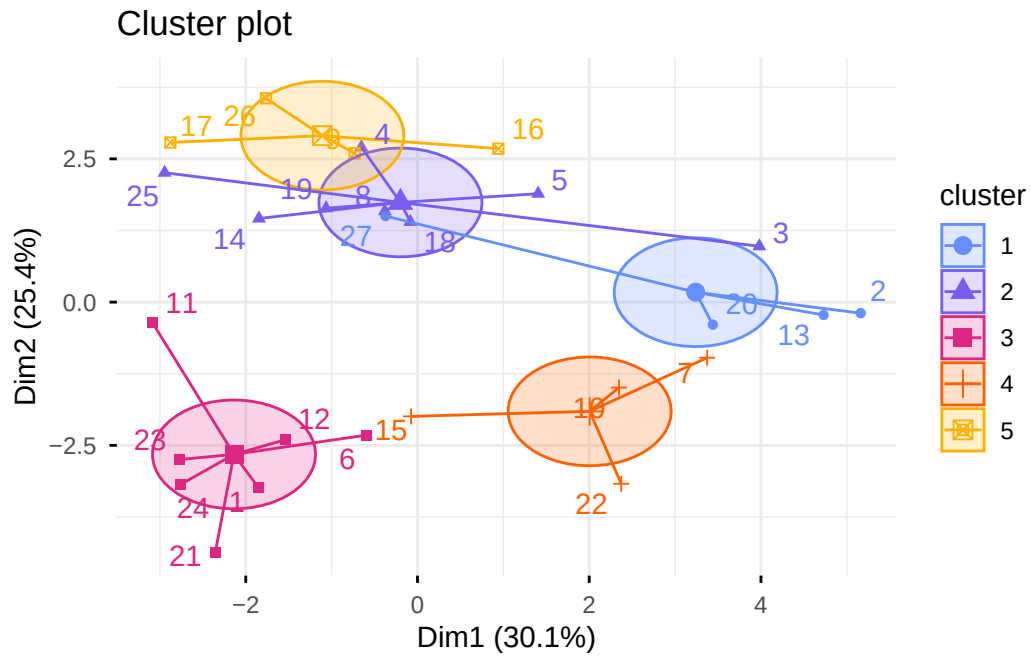


The silhouette value/metric is a measurement of the quality of a clustering. between -1 and 1

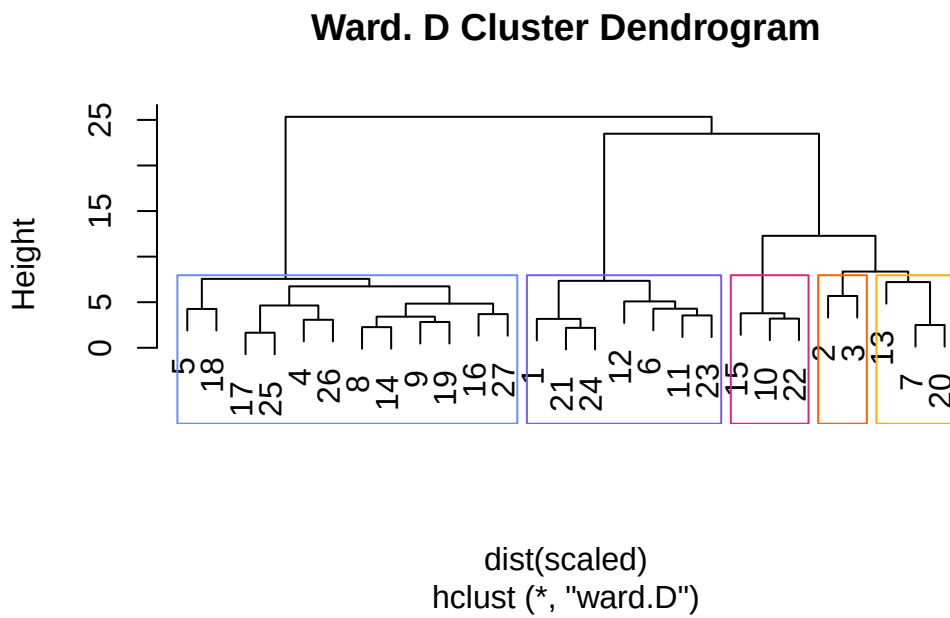
Reports optimal silhouette width as 5 with 0.31014624, indicating some weak clustering. Using complete data set or any of the other variations does not affect the suggested average silhouette. Stick with 5. ***6 is also very close.

Due to the weak clustering, we opted to perform more cluster analyses to determine if this was a good fit.

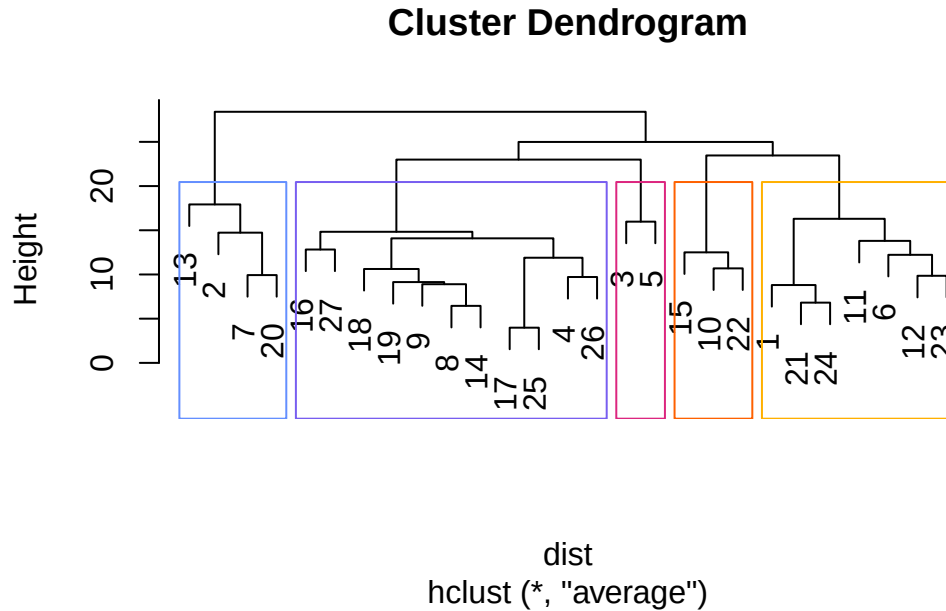
3.3 Cluster Analysis



for the clustering below based on the PAM we will be taking $k = 5$



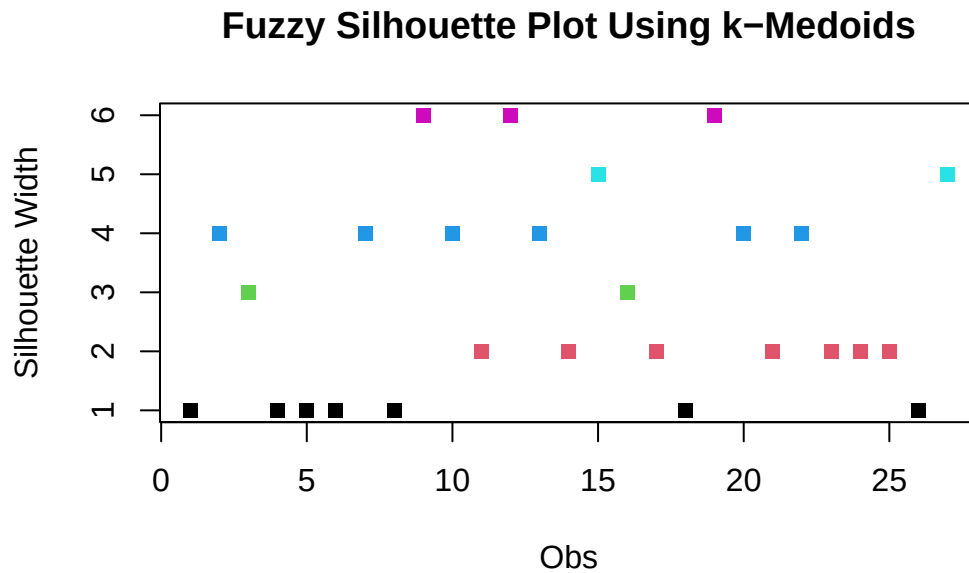
However we may not want to use wards method as ... {ALICE} ... hence Manhattan distance cluster may be a better cluster to consider as it more robust to outliers then Euclidian.



Something interesting top note is that for both cluster options they both cluster sample's 15, 10, and 22 this could be because of ...

3.4 Fuzzy Analysis lastly we decided to perform a Fuzzy analysis of the clean data as another form of clustering ... {ALICE}

The default value `alpha=1` has been set



4. Results

5. Discussion

This paper was a further analysis on the paper (insert paper and brief summary here smile), first two authors are the same.

In the initial analysis of the orichalcum ingots, X-Ray Fluorescence (XRF) spectroscopy via a portable spectrometer was used. This study includes that data and further investigates the composition of each ingot. A micro-well was drilled into each ingot and the extracted alloy was

Solved the mystery of the 2 missing ingots, they were on loan to a museum for display purposes.

What makes this discovery so important?

There has never been a discovery of this many ingots in one place. There was evidence of consistent casting processes and 3 distinct groupings of ingots were defined from this veritable hoard. This, again, is important because it provides evidence of multiple ingots of a scarce and highly prized alloy being manufactured in one place at one time.

What is Orichalcum?

Defined as a copper alloy with zinc content as the principal components, and containing a widely varying quantity of lead and tin among other trace elements. There is no consistent proportion to the amount of copper and zinc found, but the overall composition and relative amounts of principal elemental components can be used to deduce the rough date and origin of a sample. The modern equivalent is brass.

What were the limitations of this study?

A study on a sample group this large has never been done before. Also, this dataset is not very large. - love this sentence 10/10.

Why did they wait to do further testing that would provide more information on the composition and provide more accurate data?

long story short, this is a historical find and it would have taken time to gain approval to do any invasive testing that would harm or alter the ingots in any way. XRF is a portable, non-invasive method of getting accurate readings within a certain range of elements (find that bit again). The ICP-OE and ICP-MS chemometric techniques require a small amount of material to be sacrificed in order to obtain a full range of the elemental composition of each ingot.

What further research could be done? Different statistical analysis?

The paper proposes further research aimed to verify locations of ores and workshops where these, and other orichalcum, were manufactured. Providing a more fleshed out history of ancient metallurgy and possibly leading to more significant archaeological finds.

Other studies (find the papers, idiot) have classified groupings to orichalcum artifacts by their shape, composition, etc. Which has provided a map of manufacturing locations from Greece to the Caucasus.

Findings were dated to ~600BCE, the approximate age of the ship, and by the pottery also found in the wreck.

Overall similarities suggest a rudimentary mold production, likely a hole dug in the ground and molten alloy was poured in the indentation to solidify. The morphology of the found ingots suggested cast-diverse technologies. Impurities are attributed to a separation process that occurred during the fusion (CITATION) of metals to create the alloy.

Three distinctive families of ingots characterised by the graphical location of the furnace, stages of technology, and raw materials used in the smelting process.

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